

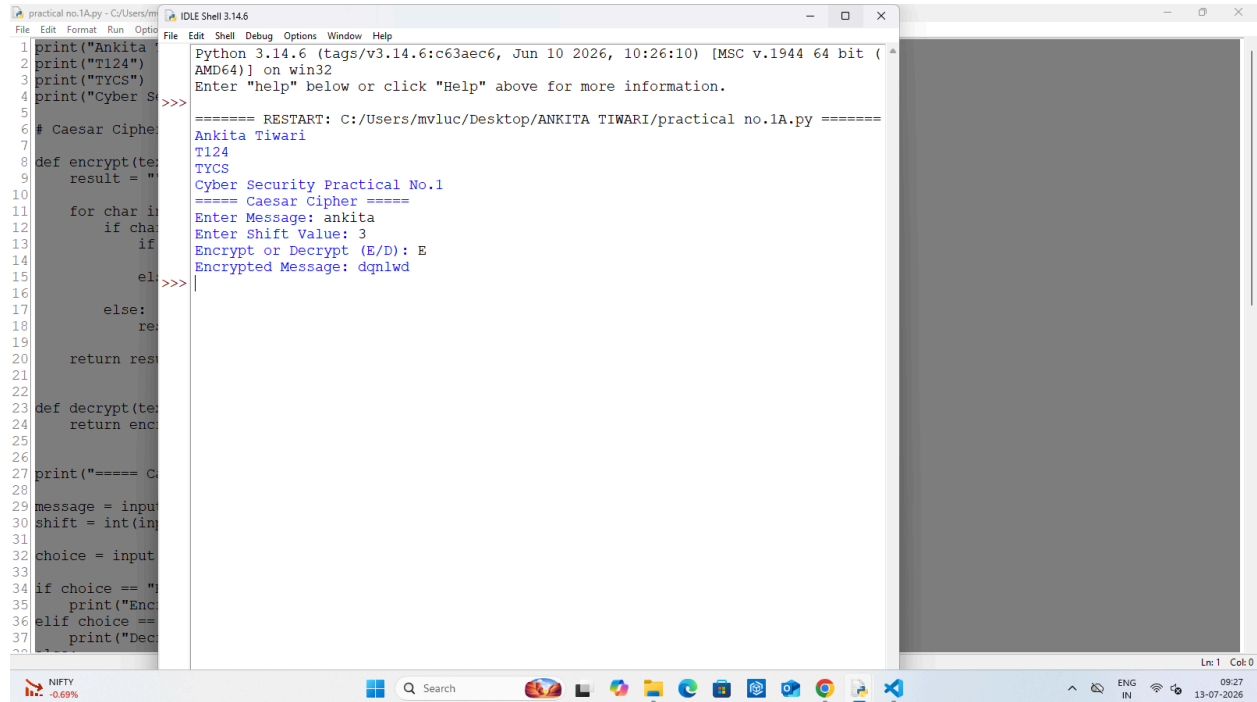
MVLU COLLEGE

1.A

Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

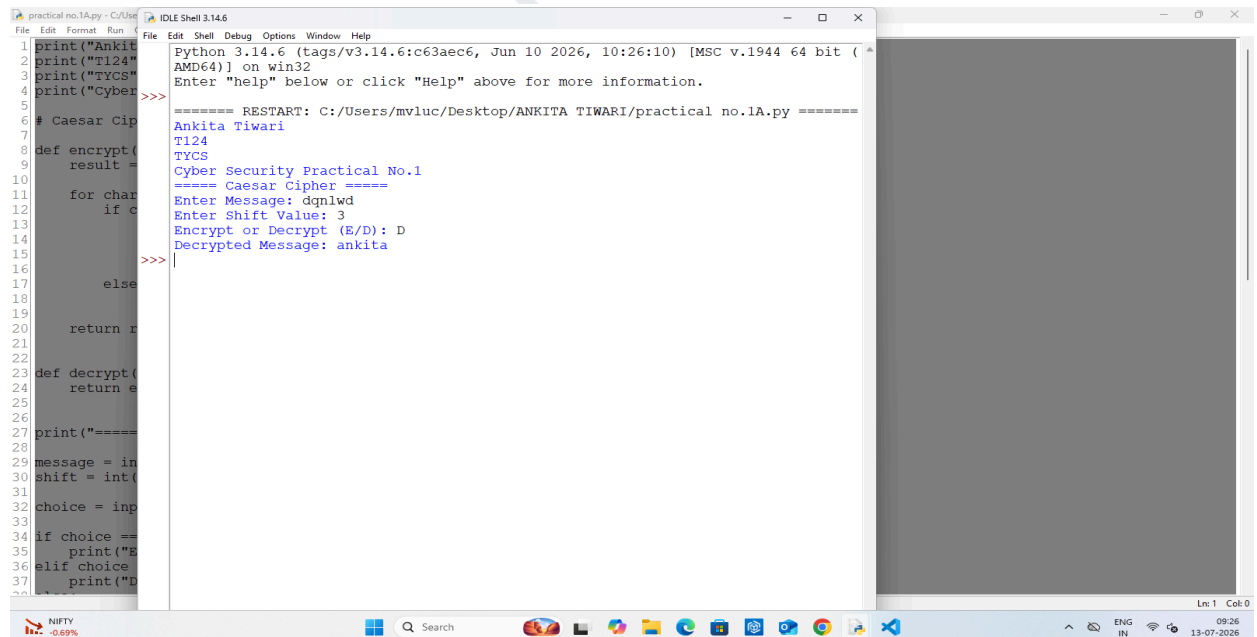
CLI METHODS:

OUTPUT:



```
practical no.1A.py - C:/Users/mvlu/Desktop/ANKITA TIWARI/practical no.1A.py
1 print("Ankita")
2 print("T124")
3 print("TYCS")
4 print("Cyber Security Practical No.1")
5
6 # Caesar Cipher
7
8 def encrypt(text, shift):
9     result = ""
10    for char in text:
11        if char.isalpha():
12            if char.isupper():
13                new_char = chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
14            else:
15                new_char = chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
16            result += new_char
17        else:
18            result += char
19    return result
20
21 def decrypt(text, shift):
22     return encrypt(text, -shift)
23
24 print("==== Caesar Cipher ====")
25 message = input("Enter Message: ")
26 shift = int(input("Enter Shift Value: "))
27 choice = input("Encrypt or Decrypt (E/D): ")
28
29 if choice == "E":
30     print("Encrypted Message: ", encrypt(message, shift))
31 elif choice == "D":
32     print("Decrypted Message: ", decrypt(message, shift))
33 else:
34     print("Invalid choice")
35
36 if __name__ == "__main__":
37     main()
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```

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
===== RESTART: C:/Users/mvlu/Desktop/ANKITA TIWARI/practical no.1A.py =====
Ankita Tiwari
T124
TYCS
Cyber Security Practical No.1
==== Caesar Cipher ====
Enter Message: ankita
Enter Shift Value: 3
Encrypt or Decrypt (E/D): E
Encrypted Message: dqnlwd
>>>
```



```
practical no.1A.py - C:/Users/mvlu/Desktop/ANKITA TIWARI/practical no.1A.py
1 print("Ankita")
2 print("T124")
3 print("TYCS")
4 print("Cyber Security Practical No.1")
5
6 # Caesar Cipher
7
8 def encrypt(text, shift):
9     result = ""
10    for char in text:
11        if char.isalpha():
12            if char.isupper():
13                new_char = chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
14            else:
15                new_char = chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
16            result += new_char
17        else:
18            result += char
19    return result
20
21 def decrypt(text, shift):
22     return encrypt(text, -shift)
23
24 print("==== Caesar Cipher ====")
25 message = input("Enter Message: ")
26 shift = int(input("Enter Shift Value: "))
27 choice = input("Encrypt or Decrypt (E/D): ")
28
29 if choice == "E":
30     print("Encrypted Message: ", encrypt(message, shift))
31 elif choice == "D":
32     print("Decrypted Message: ", decrypt(message, shift))
33 else:
34     print("Invalid choice")
35
36 if __name__ == "__main__":
37     main()
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```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
===== RESTART: C:/Users/mvlu/Desktop/ANKITA TIWARI/practical no.1A.py =====
Ankita Tiwari
T124
TYCS
Cyber Security Practical No.1
==== Caesar Cipher ====
Enter Message: dqnlwd
Enter Shift Value: 3
Encrypt or Decrypt (E/D): D
Decrypted Message: ankita
>>>
```

ANKITA TIWARI

T124

TYCS

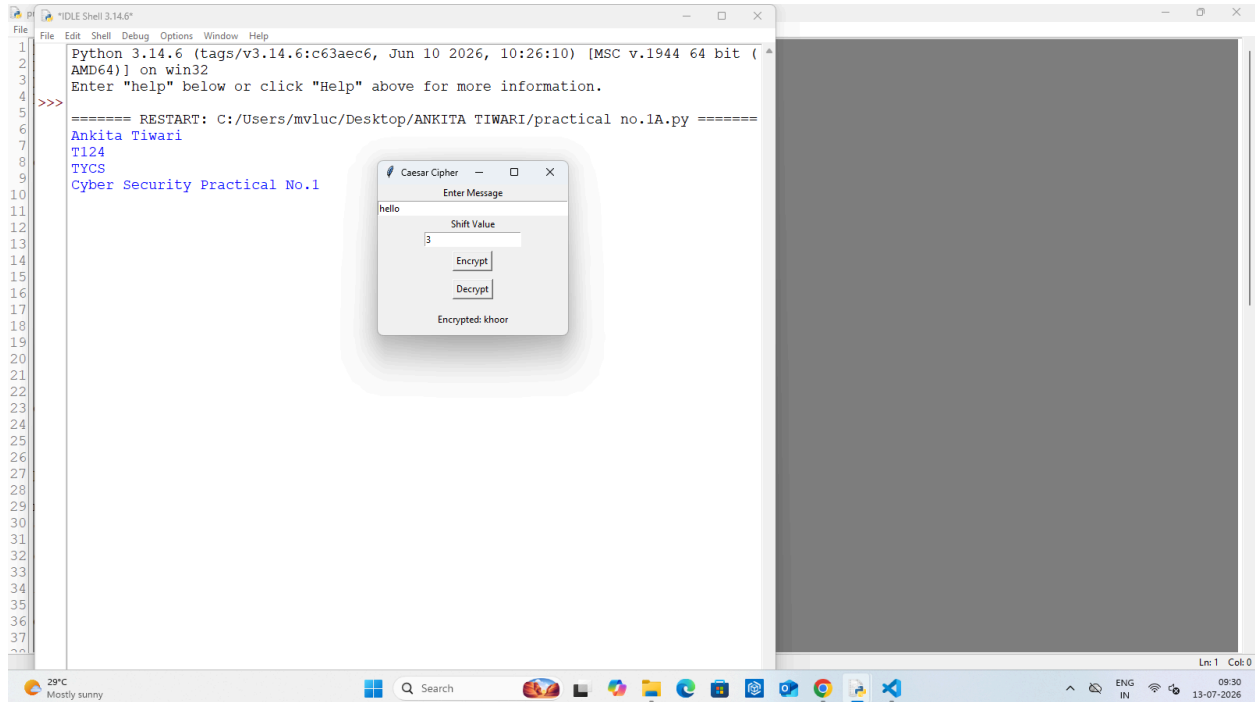
CYBER SECURITY PRACTICAL NO.1

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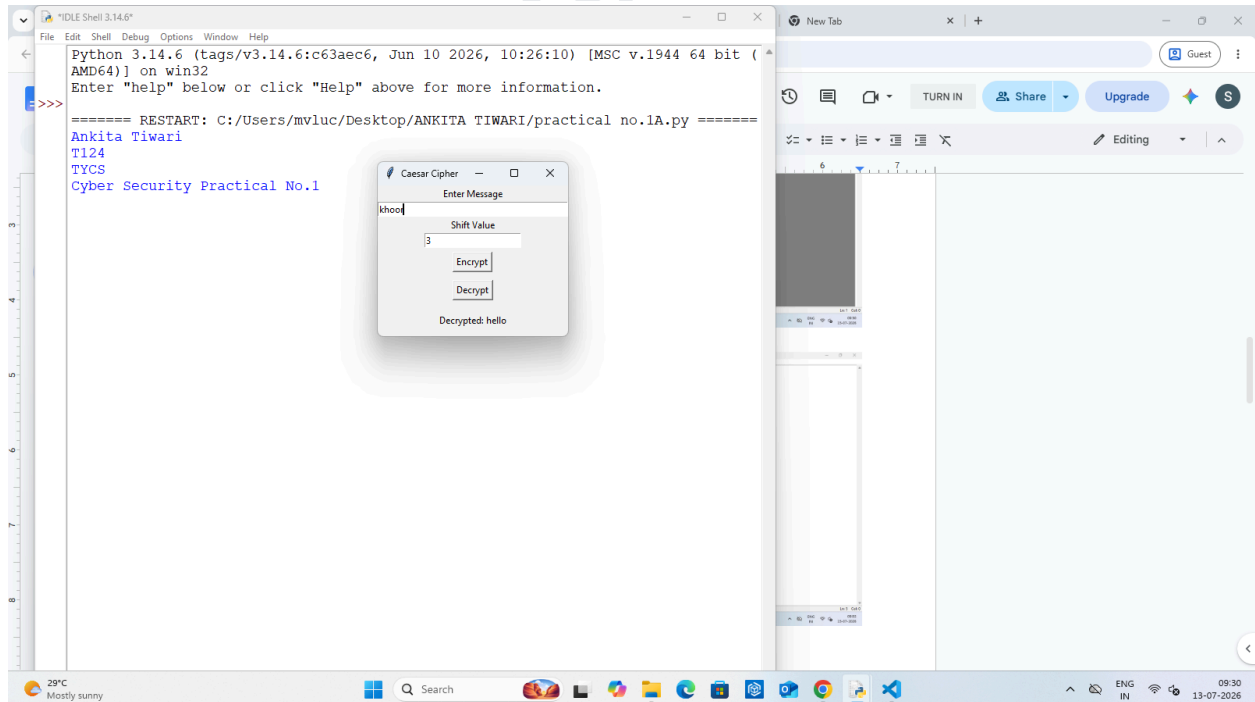
GUI METHOD:

OUTPUT:

ENCRYPT:



DECRYPT:



ANKITA TIWARI

T124

TYCS

CYBER SECURITY PRACTICAL NO.1

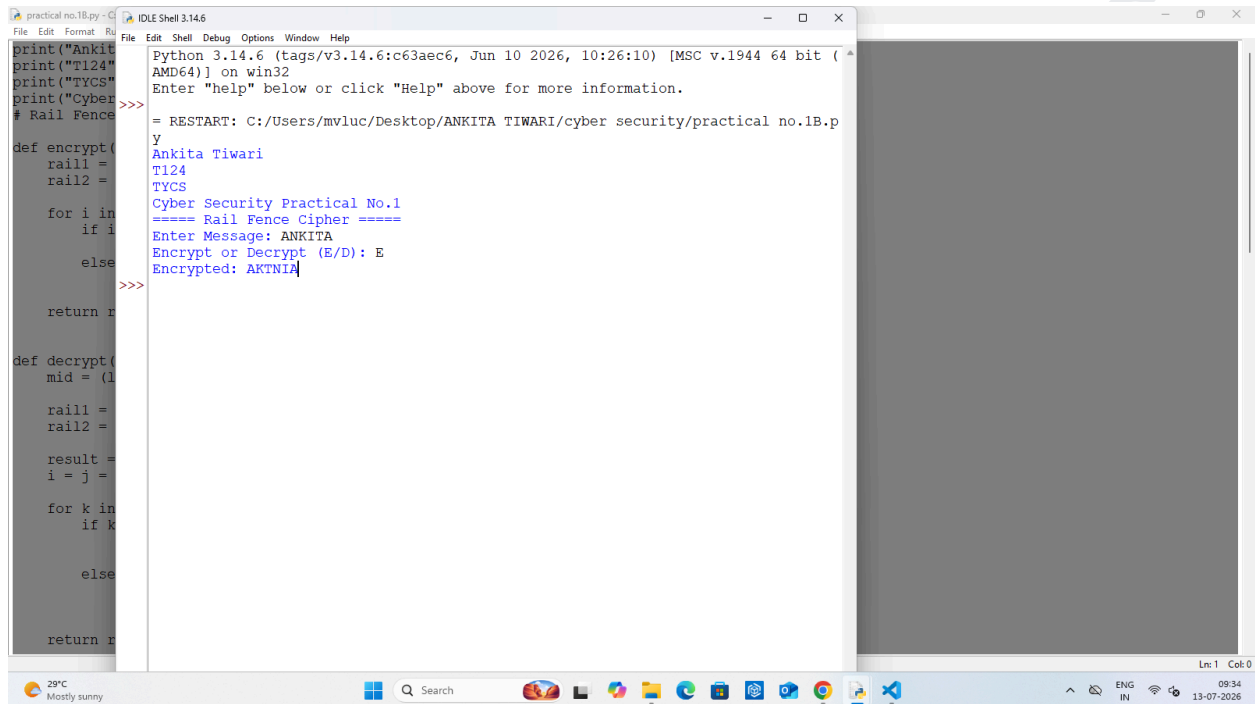
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1. B

Design and implement algorithms to encrypt and decrypt messages using transposition techniques

CLI METHODS:

OUTPUT:



The screenshot shows a Python IDE window titled 'practical no.1B.py' with a Python 3.14.6 shell. The code implements a Rail Fence cipher. The output of the program is as follows:

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/mvluc/Desktop/ANKITA TIWARI/cyber security/practical no.1B.p
Y
Ankita Tiwari
T124
TYCS
Cyber Security Practical No.1
===== Rail Fence Cipher =====
Enter Message: ANKITA
Encrypt or Decrypt (E/D): E
Encrypted: AKTNIA
```

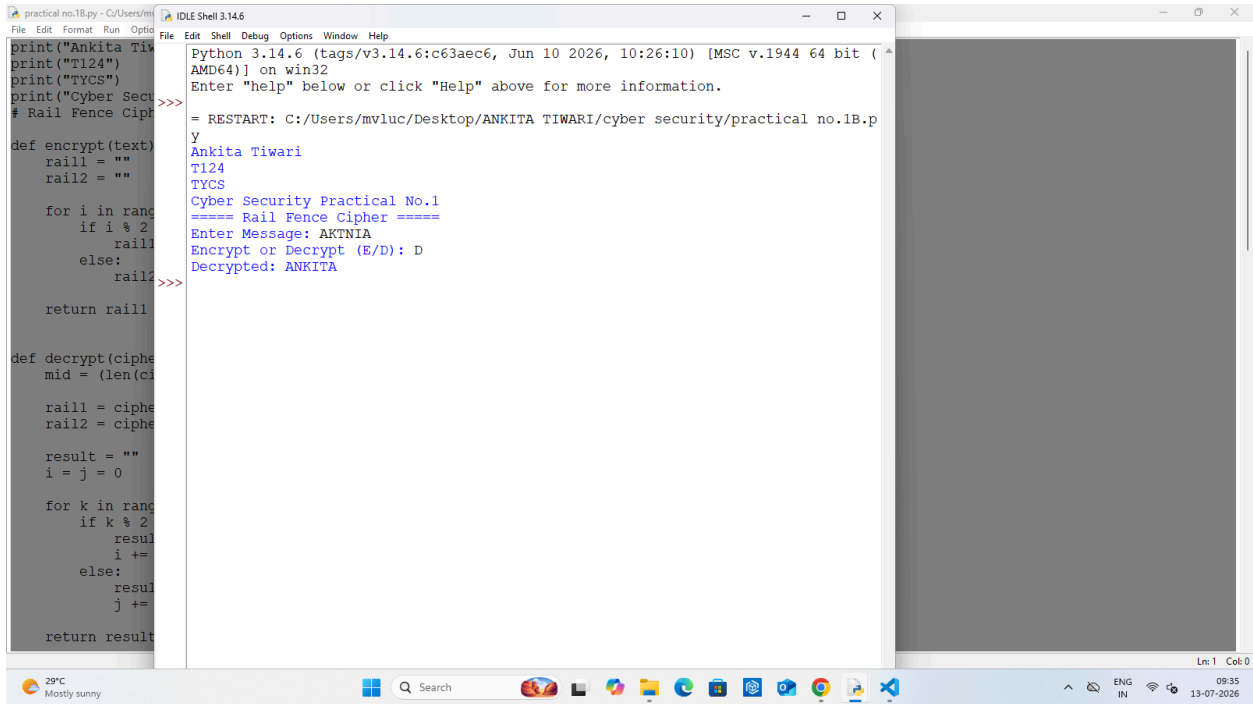
ANKITA TIWARI

T124

TYCS

CYBER SECURITY PRACTICAL NO.1

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```
practical no.1B.py - C:/Users/mvlu/Desktop/ANKITA TIWARI/cyber security/practical no.1B.py
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/mvlu/Desktop/ANKITA TIWARI/cyber security/practical no.1B.py
Ankita Tiwari
T124
TYCS
Cyber Security Practical No.1
===== Rail Fence Cipher =====
Enter Message: AKTNIA
Encrypt or Decrypt (E/D): D
Decrypted: ANKITA

def encrypt(text):
    rail1 = ""
    rail2 = ""

    for i in range(len(text)):
        if i % 2 == 0:
            rail1 += text[i]
        else:
            rail2 += text[i]

    return rail1 + rail2

def decrypt(ciphertext):
    mid = len(ciphertext) // 2
    rail1 = ciphertext[:mid]
    rail2 = ciphertext[mid:]

    result = ""
    i = j = 0

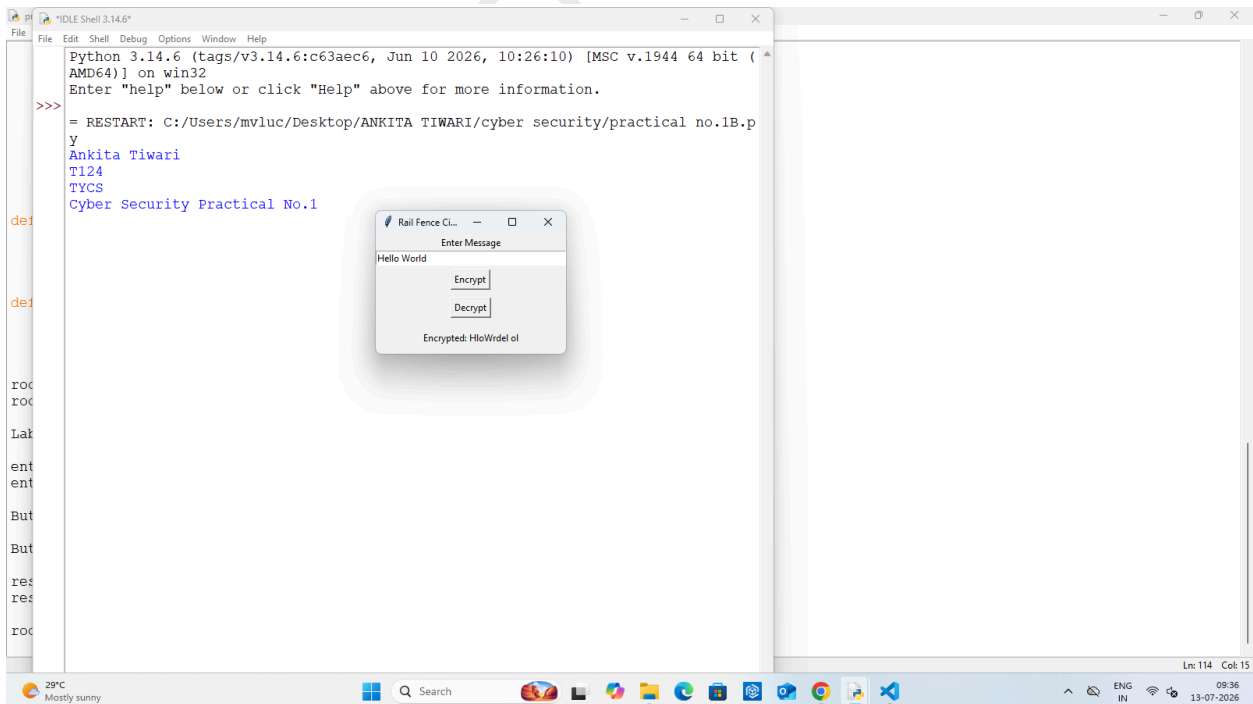
    for k in range(len(ciphertext)):
        if k % 2 == 0:
            result += rail1[i]
            i += 1
        else:
            result += rail2[j]
            j += 1

    return result
```

GUI METHOD:

OUTPUT:

ENCRYPT:



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/mvlu/Desktop/ANKITA TIWARI/cyber security/practical no.1B.py
Ankita Tiwari
T124
TYCS
Cyber Security Practical No.1

def encrypt(text):
    rail1 = ""
    rail2 = ""

    for i in range(len(text)):
        if i % 2 == 0:
            rail1 += text[i]
        else:
            rail2 += text[i]

    return rail1 + rail2

def decrypt(ciphertext):
    mid = len(ciphertext) // 2
    rail1 = ciphertext[:mid]
    rail2 = ciphertext[mid:]

    result = ""
    i = j = 0

    for k in range(len(ciphertext)):
        if k % 2 == 0:
            result += rail1[i]
            i += 1
        else:
            result += rail2[j]
            j += 1

    return result
```

Rail Fence Cipher

Enter Message

Hello World

Encrypt

Decrypt

Encrypted: HloWrld ol

ANKITA TIWARI

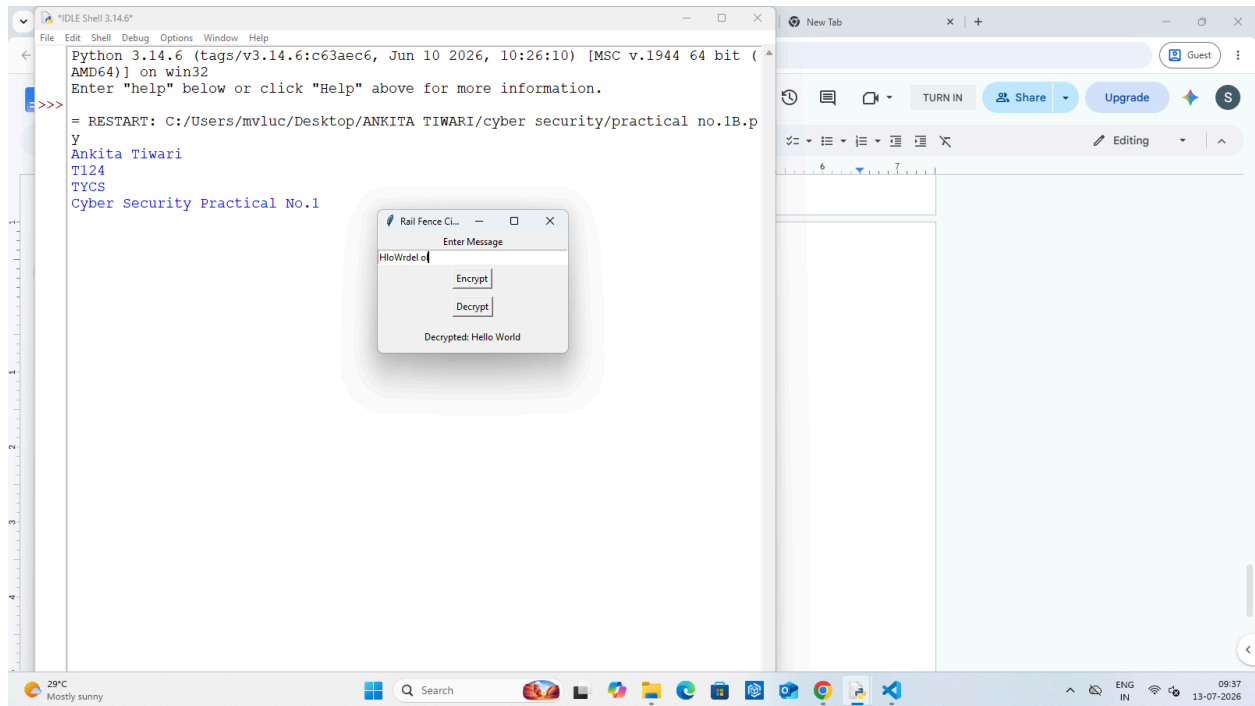
T124

TYCS

CYBER SECURITY PRACTICAL NO.1

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DECRYPT:



ANKITA TIWARI

T124

TYCS

CYBER SECURITY PRACTICAL NO.1